

EXPERIMENT 30:

```
trish@Trisha-PC: ~  
GNU nano 8.7.1 Exp30.cpp  
#include <systemc.h>  
  
SC_MODULE(rtos_scheduler) {  
    void task1() {  
        while(true) {  
            cout << "Task1 Running at "  
                << sc_time_stamp() << endl;  
            wait(1, SC_SEC);  
        }  
    }  
  
    void task2() {  
        while(true) {  
            cout << "Task2 Running at "  
                << sc_time_stamp() << endl;  
            wait(1, SC_SEC);  
        }  
    }  
  
    SC_CTOR(rtos_scheduler) {  
        SC_THREAD(task1);  
        SC_THREAD(task2);  
    }  
};  
  
int sc_main(int argc, char* argv[]) {  
  
    ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^I Execute    ^C Location   M-U Undo      M-A Set Mark  
    ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_/ Go To Line M-E Redo      M-6 Copy
```

```
trish@Trisha-PC:~$ nano Exp30.cpp  
trish@Trisha-PC:~$ g++ -o Exp30 Exp30.cpp -lsystemc  
trish@Trisha-PC:~$ ./Exp30  
  
SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46  
Copyright (c) 1996-2025 by all Contributors,  
ALL RIGHTS RESERVED  
Task1 Running at 0 s  
Task2 Running at 0 s  
Task1 Running at 1 s  
Task2 Running at 1 s  
Task1 Running at 2 s  
Task2 Running at 2 s  
Task1 Running at 3 s  
Task2 Running at 3 s  
Task1 Running at 4 s  
Task2 Running at 4 s  
Task1 Running at 5 s  
Task2 Running at 5 s  
trish@Trisha-PC:~$ |
```